Physical Chemistry Cume

Sept 7th 2018

1) For the reversible reaction

$$A + B \rightleftharpoons C + D$$

Various initial rate measurements were run starting with A and B only, and starting with C and D only:

[A] / M	[B] / M	Initial Rate / M s^{-1}
0.667	1.23	1.10×10^{-5}
4.00	1.23	6.60×10^{-5}
4.00	2.25	6.60×10^{-5}

[C] / M	[D] / M	Initial Rate / M s^{-1}
3.00	1.00	7.875×10^{-7}
3.00	1.65	1.260×10^{-6}
1.00	1.65	1.260×10^{-6}

Determine the equilibrium constant for the reaction

2) For the following reversible, first order reaction

$$A \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} B$$

One flask contains a solution of only A at a concentration of 1.6 M, and another flask contains a solution of only B at a concentration of 1.0 M. Equal volumes of each solution were rapidly mixed and the change in the concentration of A was measured as a function of time as follows:

t / min	0.5	1	1.5	2	3	4	5	10
[A] / M	0.670	0.600	0.563	0.543	0.527	0.522	0.520	0.520

Determine k_1 and k_{-1}